

Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Decision Making and Conditional Statements in C Programming
Lab Number	04
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab 04: Decision Making and Conditional Statements in C Programming

Objective

To learn and implement decision-making in C programming using if, if-else, and nested if-else statements for solving real-world logical problems.

Software Requirements

- Code::Blocks IDE / Dev-C++
- GCC Compiler
- Windows Operating System

Experiments Performed

In-Lab Tasks

1. Determine whether an entered integer is even or odd.
2. Determine whether a driver is eligible for insurance based on age, gender, and marital status.

Post-Lab Tasks

1. Check whether a given year is a leap year.
2. Determine scholarship eligibility using percentage, NAT score, and orphan status.
3. Calculate library fines based on the number of late days and determine membership cancellation.
4. Identify whether an entered character is an alphabet, digit, or special character using ASCII values.

Learning Outcomes

After completing this lab, the student will be able to:

- Use if, if-else, and nested if-else statements.
- Implement logical conditions using relational and logical operators.

- Solve decision-based programming problems.
- Validate user input and perform conditional checks.
- Develop simple real-world applications using conditional statements.

In lab Tasks:

Task 1:

Ask the user to enter an integer. Check whether the entered integer is even or odd.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int s;
    printf("Enter an Integer:\n");
    scanf("%d", &s);
    if(s % 2==0)
        printf("\nNumber is Even\n");
    else
    {
        printf("\nNumber is Odd\n");
    }
    return 0;
}
```

Output:

Task 2:

A company insures its driver in the following cases

- if the driver is married.
- If the driver is unmarried, male& above 30 years of age.
- If the driver is unmarried, female& above 25 years age?

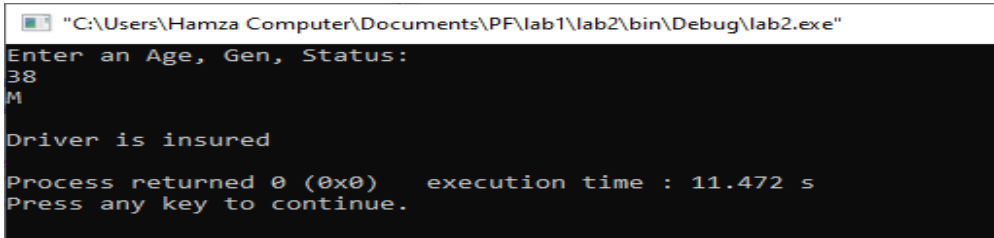
Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    char Gen,Status;
    int Age;
    printf("Enter an Age, Gen, Status:\n");
    scanf("%d%c%c",&Age,&Gen,&Status);

    if(Status=='M')
        printf("\nDriver is insured\n");
    else
    {
        if (Gen=='M')
        {
            if (Age>30)
                printf("\nDriver is insured\n");
            else
                printf("\nDriver is not insured\n");
        }
        else
        {
            if (Age>25)
                printf("Driver is insured");
            else
                printf("\nDriver is not insured\n");
        }
    }
    return 0;
```

```
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter an Age, Gen, Status:
38
M
Driver is insured
Process returned 0 (0x0)   execution time : 11.472 s
Press any key to continue.
```

Post Lab Tasks:

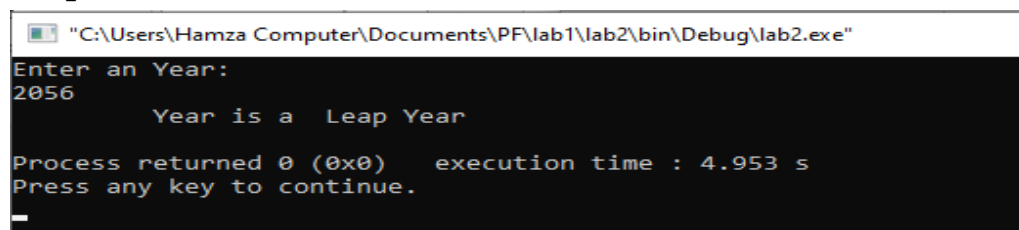
Task 1:

Any year is input through the keyboard. Write a program to determine whether the year is leap or not.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int Year;
    printf("Enter an Year:\n");
    scanf("%d", &Year);
    if(Year % 4==0)
        printf("\t Year is a Leap Year\n");
    else
        printf("\t Year is not a Leap Year\n");
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter an Year:
2056
Year is a Leap Year
Process returned 0 (0x0)   execution time : 4.953 s
Press any key to continue.
```

Task 2:

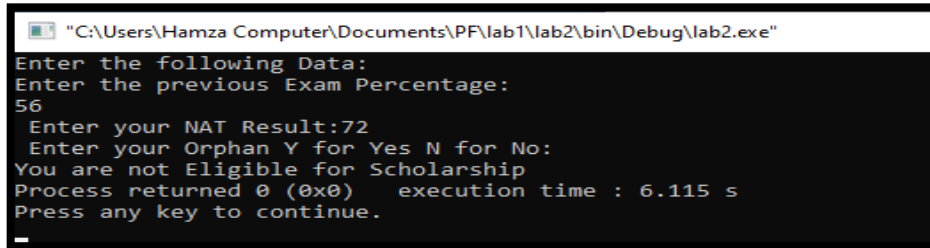
Determine scholarship eligibility using percentage, NAT score, and orphan status.

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    float Per, NAT;
    char Orphan, Y, N;
    printf("Enter the following Data:\n");
    printf("Enter the previous Exam Percentage:\n");
    scanf("%f", &Per);
    printf(" Enter your NAT Result:");
    scanf("%f", &NAT);
    printf(" Enter your Orphan Y for Yes N for No:");
    scanf("%c", &Orphan);
    if(Per>90)
    {
        printf("\nYou are eligible for Scholarship");
    }
    else
        if (Per>70 && Orphan=='Y' && NAT>80)
        {
            printf("\nYou are eligible for Scholarship");
        }
    else
        if (Per>80 && Orphan=='Y' && NAT>70)
        {
            printf("\nYou are Eligible for Scholarship");
        }
    else
    {
        printf("\nYou are not Eligible for Scholarship");
    }
    return 0;
}
```

```
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
Enter the following Data:
Enter the previous Exam Percentage:
56
Enter your NAT Result:72
Enter your Orphan Y for Yes N for No:
You are not Eligible for Scholarship
Process returned 0 (0x0)   execution time : 6.115 s
Press any key to continue.
```

Task 3:

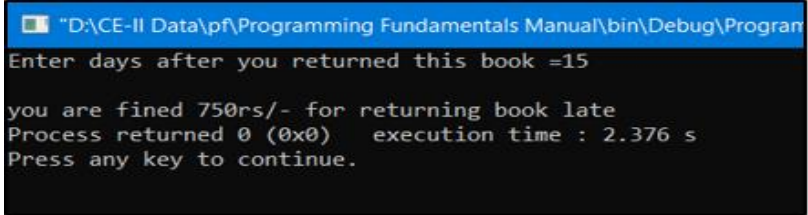
A library charges a fine for every book returned late. For first five days the fine is 5 rupee per day, for 6-10 days fine is 10 rupee per day and above 10 days fine is 50 rupee per day. If someone returns book after 30 days then his\her membership is cancelled. Write a program to accept number of days the member is late to return the book and display the fine or appropriate message?

Code:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int Day, Fine;
    Day=0;
    Fine=0;
    printf("Enter Days After You Returned this Book:\n");
    scanf("%d", &Day);
    if(Day==0);
        printf("No Fine");
    else
        if(Day<=5 &&Day>=1)
            Fine = 5*Day;
        printf("\n You are Fined  with %Rupees for Returning Book late", Fine);
    else
        if(Day>=6 &&Day<=10)
            Fine = 10*Day;
        printf("\nYou are fined with %Rupees for returning Book late", Fine);
    else
        if(Day>=11 &&Day>=30)
```

```
Fine = 50*Day;
printf("\nYou are fined with %Rupees for returning Book late", Fine);
printf("\nYour Membership has been canceled Bcz You returned Book late by more than 30
Days");
return 0;
}
```

Output:



```
"D:\CE-II Data\pf\Programming Fundamentals Manual\bin\Debug\Program
Enter days after you returned this book =15
you are fined 750rs/- for returning book late
Process returned 0 (0x0)   execution time : 2.376 s
Press any key to continue.
```

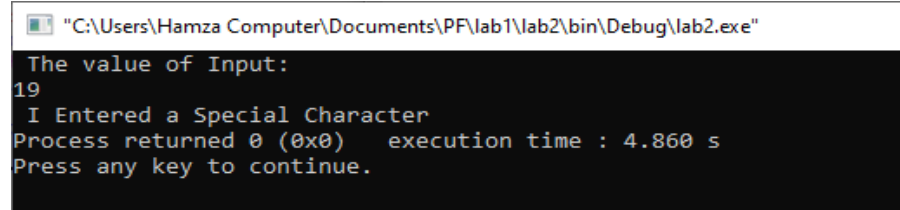
Task 4:

Write a C program to check alphabet, digit or special character using ASCII?

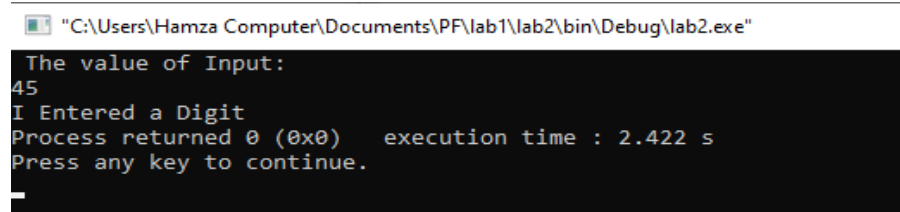
Code:

```
#include <stdlib.h>
int main()
{
    char C;
    int D;
    printf(" The value of Input:\n");
    scanf("%d", &D);
    D==C;
    if(D<=122 && D>=97 || D>65 && D<90)
        printf("I Entered an Alphabet:\n");
    else
        if(D>=32 && D<=47 || D>=58 && D<=64 || D>=91 && D<=96 || D>=123 && D<=126)
            printf(" I Entered a Special Character");
        else
            printf("I Entered a Digit");
    return 0;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
The value of Input:
19
I Entered a Special Character
Process returned 0 (0x0)   execution time : 4.860 s
Press any key to continue.
```



```
"C:\Users\Hamza Computer\Documents\PF\lab1\lab2\bin\Debug\lab2.exe"
The value of Input:
45
I Entered a Digit
Process returned 0 (0x0)   execution time : 2.422 s
Press any key to continue.
```